Table 1. Outline of Recommended Documentation

Software Documentation Elements	Requirements
Software Description	Software description, including overview of significant software features, functions, analyses, inputs, outputs, and hardware platforms.
Risk Management File	Risk management plan, risk assessment demonstrating that risks have been appropriately mitigated, and risk management report.
Software Requirements Specification (SRS)	SRS documentation, describing the needs or expectations for a system or software, presented in an organized format, at the software system level or subsystem level, as appropriate, and with sufficient information to understand the traceability of the information with respect to the other software documentation elements (e.g., risk management file, software design specification, system and software architecture design chart, software testing).
System and Software Architecture Design	Detailed diagrams of the modules, layers, and interfaces that comprise the device, their relationships, the data inputs/outputs and flow of data, and how users or external products (including information technology (IT) infrastructure and peripherals) interact with the system and software.
Software Design Specification (SDS)	Basic Level FDA is not recommending the SDS as part of the pre-market submission. Sponsor should document this information on the design via the DHF for the device. During premarket review, FDA may request additional information, i needed, to evaluate the safety and effectiveness of the device. Enhanced Level SDS documentation, including sufficient information that would allow FDA to understand the technical design details of how the software functions, how the software design completely and correctly implements all the requirements of the SRS, and how the software design traces to the SRS in terms of intended use, functionality, safety, and effectiveness.

Software Description

An overview of significant software features and functions, including images, flow charts, and state diagrams as needed to adequately explain the software functionality[1] should be provided. If the premarket submission is for a modified device, provide the document number of the previous submission and highlight pertinent software changes (e.g., changes that affect safety and effectiveness) since the last FDA approval or clearance.

Consider and, as applicable, provide information to address the questions below when preparing the software description. However, FDA recognizes that these questions and examples may not capture all the unique aspects of device software and encourages the inclusion of additional information that will further FDA's understanding of the device's functionality to facilitate the review of a submission.

- Software Operation
  - What is the role of the software as it relates to the intended use of the device? *Examples: software-only device, software that controls the device hardware, software application that accompanies the device for data processing.*
  - Who is the intended user of the software? *Examples: the patient, a caregiver, a healthcare professional, or a combination thereof*
  - What is the intended patient population?
    - Does the software function focus on a specific disease, condition, patient characteristic or demographic?
    - Does the software provide information that is directly applicable to a specific disease or condition?
  - If the software performs analysis of signals/patterns or images, what is the analysis methodology? *Examples: rule-based calculations, online test administration, artificial intelligence and machine learning (AI/ML), neural networks, fixed (locked) or adaptive (iterative, continuously learning) algorithms.*
  - If the device software function uses ML models trained through ML methods (i.e., is AVML-enabled, such as AI, ML, adaptive models, natural language processing (NLP), neural networks, and related approaches):
    - What methods, models, frameworks, and/or platforms were used?
    - What data (populations, samples) informed the model(s)? How, when, and where was the data collected?
    - What steps were taken to identify and address potential biases and limitations of the model(s)?
    - What materials, mechanisms, and/or approaches are used to provide transparency about the model's development, performance, and limitation(s)?

- Software Specifics[2]
  - What hardware platforms are used?
  - What software platforms are used?
    - If applicable, what hosting environments are used (e.g., hospital networks, cloud infrastructures) and for what functions (e.g., processing, storage)?
  - Does the device use OTS software?[3]
  - What is the final release version (i.e., version intended to be released to end users)? If this version is different from the documentation's version, explain the differences.
- Software Inputs and Outputs
  - What are the inputs and their format? *Examples: signals, images (specify modality), measurements (specify units), reports, questionnaires, other device data/results.*
  - Who or what provides the inputs? *Examples: patients, caregivers, healthcare professionals, technicians, sensors/attachments, signal acquisition systems, in vitro diagnostic devices, other medical devices, other non-medical products or software.*
  - What are the outputs and their format? *Examples: diagnostic information, treatment information, control signals for device hardware, images (specify modality), measurements (specify units), alarms, alerts, or reports.*
  - Who or what receives the outputs? *Examples: patients, caregivers, healthcare professionals, technicians, health records, device hardware, other medical devices, interoperable systems.*
  - Does the software impact or replace any action(s) otherwise performed manually by a health care professional, patient, caregiver, or other operator? What are the clinical workflow steps and assumptions (from beginning to end state)? *Examples: automates steps, triages patients, provides a definite diagnosis or suggests likely diagnosis for further confirmation by physician, performs or recommends specific treatment, identifies a region of interest for further review.*
  - Is the device designed to be interoperable? In other words, does the device transmit, exchange, and/or use information through an electronic interface (e.g., network, wireless) with another medical/non-medical product, system, or device? If yes, what other products does the device interface with, and what methods, standards, and specifications are used to interact and/or communicate with other medical/non-medical product, system, or device? Are the medical/non-medical products, systems, or devices networked?

[1] FDA may request additional architecture diagrams to address the cybersecurity risks associated with a device.

[2] One example approach for providing some of the identified software specifics is by providing a Software Bill of Materials (SBOM) that lists and provide details on software components including, but not limited to, commercial, open source, OTS, and manufacturer-developed software components. For some devices, information related to an SBOM may be required part of a premarket submission.

[3] Off-the-Shelf (OTS) Software - A generally available software component used by a device manufacturer for which the manufacturer cannot claim complete software lifecycle control (e.g., operating system, printer/display libraries). If a device uses OTS software, FDA may request additional information in premarket submissions. For more information, please refer to the guidance document "Guidance for Off-The-Shelf Software Use in Medical Devices."